A guide to assumptions in introductory thermodynamics

Generally, we should make only the assumptions that are necessary to solve a problem. Thermodynamics is powerful in part because it requires very few assumptions, and so applies to a very wide range of systems. As Einstein wrote[1], "A theory is the more impressive the greater the simplicity of its premises, the more different kinds of things it relates, and the more extended its area of applicability. Therefore the deep impression that classical thermodynamics made upon me."

Here is a list of assumptions we could potentially make when solving a thermodynamics homework or exam problem, with guidance on when to make each assumption. The scope of this list is the first unit of Purdue ME 200, which covers the first law for closed systems. Problems involving mass transfer (the second unit of ME 200) or entropy calculations (the third unit) may involve different assumptions.

1. **Closed system:** There is no mass transfer across the system boundary over a process. Useful consequences are that the system mass is constant and that the first law for closed systems holds. Make this assumption if you use $\Delta E = Q - W$.

2. **Rigid container:** A container's volume is constant. A useful consequences is that if the system occupies the whole container over a process, then the boundary work is zero. If the system is also closed, then specific volume is constant. Make this assumption if the problem statement calls the container 'rigid.' Your solution does not need to state this assumption.

3. **Other constant property:** A system property (such as pressure, temperature, internal energy, or enthalpy) is constant over a process. Make this assumption if the problem statement says so. Your solution does not need to state this assumption.

4. **No center-of-mass motion:** The system's center-of-mass position and velocity are constant over a process. A useful consequence is that the system's macroscopic gravitational potential energy and macroscopic kinetic energy are constant: $\Delta\text{PE} = \Delta\text{KE} = 0$, so $\Delta E = \Delta U$. Make this assumption unless the problem explicitly mentions changes to macroscopic PE or KE.

[1] Einstein, A. and Schilpp, P. (editor): Autobiographical Notes. Open Court Publishing (1979).

5. **Well-insulated wall(s):** There is no heat transfer across a container wall or walls over a process. Make this assumption if the problem statement calls the container 'well-insulated' or 'adiabatic.'

6. **Ideal gas:** A gas satisfies the ideal gas law, $pv = RT$, and its internal energy is a function of temperature only. Make this assumption if the problem statement asks you to or the problem is unsolvable without it.

7. **Incompressible substance:** A solid or liquid has constant specific volume and its internal energy is a function of temperature only. Make this assumption if the problem statement asks you to or the problem is unsolvable without it.

8. **Constant specific heat(s):** A substance's specific heats are constant over a process. Useful consequences are that an ideal gas satisfies $\Delta u = c_v \Delta T$ and $\Delta h = c_p \Delta T$, and that an incompressible substance satisfies $\Delta u = c\Delta T$ and $\Delta h = c\Delta T + v\Delta p$. Make this assumption only if the problem statement says to assume constant specific heat(s) and/or gives a single numerical value for a specific heat over a process.

9. **Quasiequilibrium process:** The system is infinitesimally close to an (internal) equilibrium state at each instant during a process. A useful consequence is that the system properties are spatially uniform throughout the process, so the scalars T and p uniquely define the system temperature and pressure distributions, respectively. Make this assumption if you

   (a) evaluate a boundary work integral over a process that is not constant-pressure (boundary work is defined as $p\Delta V$ in constant-pressure processes) or polytropic;

   (b) plot p over a process that is not constant-pressure or polytropic;

   (c) plot T over a process that is not constant-temperature or polytropic.

10. **Polytropic process:** The system pressure and volume satisfy $pV^n = C$ for some constants n and C over a quasiequilibrium process. A useful consequence is that boundary work integrals can be easily evaluated:

$$pV^n = C \implies \int_{V_1}^{V_2} p\mathrm{d}V = \begin{cases} (p_2V_2 - p_1V_1)/(1-n) & \text{if } n \neq 1 \\ p_1V_1 \ln(V_2/V_1) & \text{if } n = 1. \end{cases}$$

    Make this assumption only if the problem statement asks you to. Your solution does not need to state this assumption. All polytropic processes are quasiequilibrium by definition[2], so it is not necessary to assume that a polytropic process is quasiequilibrium.

[2] Moran *et al*:, Fundamentals of Engineering Thermodynamics 9th edition (2018) p. 33: "A quasiequilibrium process described by pV^n = constant, or pv^n = constant, where n is a constant, is called a polytropic process."